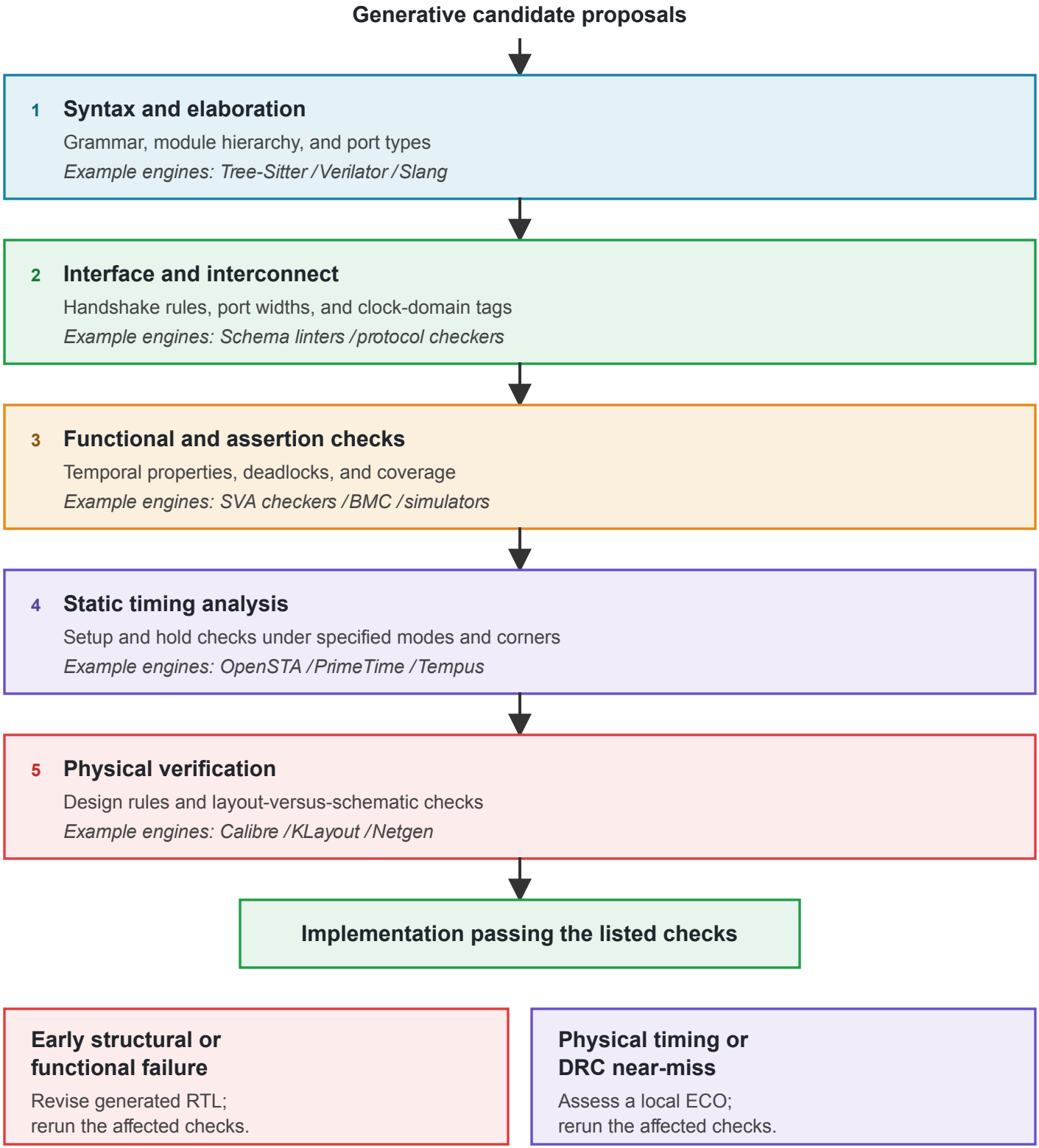

Generative candidate proposals
1 Syntax and elaboration
Grammar, module hierarchy, and port types
Example engines: Tree-Sitter / Verilator / Slang
2 Interface and interconnect
Handshake rules, port widths, and clock-domain tags
Example engines: Schema linters / protocol checkers
3 Functional and assertion checks
Temporal properties, deadlocks, and coverage
Example engines: SVA checkers / BMC / simulators
4 Static timing analysis
Setup and hold checks under specified modes and corners
Example engines: OpenSTA / PrimeTime / Tempus
5 Physical verification
Design rules and layout-versus-schematic checks
Example engines: Calibre / KLayout / Netgen
Implementation passing the listed checks
Early structural or functional failure
Revise generated RTL; rerun the affected checks.
Physical timing or DRC near-miss
Assess a local ECO; rerun the affected checks.